Soap Bubbles

01 What is a soap bubble made of?



A watery soap film wraps a pocket of air.

Look closer. Find the thin shiny wall of the bubble, and the air space inside.

A little explanation. A soap bubble is a very thin film of water mixed with soap, wrapped around a pocket of air. Soap molecules help the watery film hold together a little longer than plain water would. The wall is so thin you can see through it, yet it still encloses the air.

Try this. Dip a wand and hold up a flat soap film (not a bubble yet). Look at the film edge-on if you can, then blow gently to trap air inside.

02 Why do bubble colors swirl and shift?



Light and a thinning film make colors shift.

Look closer. Find places on the bubble where stripes of color move and change.

A little explanation. Light reflects from the front and back of the thin soap film. Those reflections can add or cancel, so some colors look bright while others fade. As water drains or evaporates, the film gets thinner, and the color pattern shifts. Scientists call this thin-film interference.

Try this. Watch one bubble without poking it. Name two colors you see, then notice whether they move toward the top or bottom as time passes.

03 Why are free-floating bubbles round?



Surface tension pulls a free bubble into a sphere.

Look closer. Compare a free bubble in the air with any stretched film still on a wand.

A little explanation. Surface tension pulls on the soap film from every direction. For a given amount of air inside, a sphere uses the least film area, so a free bubble settles into a round shape. On a wand, the frame can stretch the film into other shapes until the air is trapped.

Try this. Blow one bubble and watch it from the side. Sketch the round outline, then write one sentence about the pull of the film.

04 Why does a bubble pop?



When the film breaks, the bubble is gone.

Look closer. Find a dark or dull patch on a bubble just before it vanishes, if you can.

A little explanation. A bubble pops when its film breaks. Touching it, dusty air, or a dry spot can open a hole. Even without a poke, water leaves the film as it drains and evaporates, so a place can get too thin to hold. Some recipes last longer, but every ordinary soap bubble still has a short life.

Try this. Time how long two bubbles last in still air. Change only one thing (shade versus sun, or a gentle cover) and compare. Do not aim bubbles at faces or eyes.